Ihor Kendiukhov
Institute of Medical Genetics and Applied Genomics
University of Tübingen
Tübingen, Germany
ihor.kendiukhov@student.uni-tuebingen.de

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The Editors
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Dear Editors,

I am submitting the manuscript **"How Faithful Is the Circuit You Found? Post-Selection-Valid Confidence Bounds for Mechanistic Interpretability"** for consideration as an *Article* in *Neural Networks*. I suggest the **Mathematical and Computational Analysis** section; **Learning Systems** would also be appropriate if the editors prefer it.

What the paper is about. Work that reverse-engineers a trained network into an interpretable circuit reports a faithfulness score: how much of the network's behaviour survives when everything outside the proposed circuit is intervened on. Almost always, that score is the largest of many — the circuit was chosen by maximising the score on the very interventions later used to report it — and it is accompanied by a standard error computed as though no search had taken place. This is the winner's curse, and it makes the field's headline numbers optimistic by an amount nobody has measured.

This manuscript quantifies the problem and fixes it. It formalises graded faithfulness as a population quantity and circuit search as a selection rule, proves that the inflation of a selected score is of order $\sigma\sqrt{\log|\mathcal{C}|/n}$ in the size of the searched class and the number of interventions, and supplies five lower confidence bounds that remain valid however the circuit was chosen. Coverage is then *measured* rather than assumed, on five small transformers whose algorithms are known by construction (two of them with a correct circuit planted by interchange intervention training, so that the truth is available exactly) and on the indirect-object-identification circuit of GPT-2 small.

Why it may interest this readership.

- It is about neural networks specifically, not statistics in general. Three of the corrections had to be re-derived because faithfulness scores are binary, because the most widely used metric is a ratio of means rather than a mean, and because circuit classes are power sets of network components. In particular, the textbook multiplier bootstrap *under-covers* for binary faithfulness scores — 58% instead of 95% in our simulations — and we show why and what to use instead.
- The empirical findings are actionable. A nominally 95% interval on a searched circuit was correct 0.6% of the time in the controlled setting and 59% of the time on GPT-2; the valid alternative costs only 0.007 of certified faithfulness. Separately, we show that prompt *templates*, not prompts, are the unit of replication, so that generating more prompts from the same

templates does not buy the evidence practitioners assume it does.

- The paper ends with a six-item reporting protocol that adds one column to a results table and no additional model evaluations.

Reproducibility. All code, the cached per-instance score matrices, the analysis outputs and the manuscript source are public at `https://github.com/Kendiukhov/post-selection-faithfulness`. Every statistical result in the paper can be recomputed from the committed matrices without a GPU; the full pipeline, including the GPT-2 passes, runs in about five GPU-hours on a single consumer accelerator. The library ships with a test suite whose Monte-Carlo checks verify the coverage of every bound.

Declarations. The manuscript is original, is not under consideration elsewhere, and has not been published before in any form, including as a conference paper. There are no competing interests and no funding to declare. I am the sole author and take full responsibility for the content, including for the use of an AI assistant in implementing the software and drafting the text, which is disclosed in the manuscript in accordance with Elsevier policy.

I would be grateful for the editors' consideration, and I am happy to provide any further material the review process requires.

Yours sincerely,

Ihor Kendiukhov